

Math & Programming Camp

2026

Department of Government, Georgetown University

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Dates: Monday, August 17 – Friday, August 21, 2026

Time: 9:00 AM – 4:00 PM (Friday extends to 4:30 PM)

Location: ICC 450 (Mon – Thu) · ICC 662 (Fri)

Course links:

To be added.

Course description: Math & Programming Camp is designed to refresh mathematical concepts and introduce statistical programming that Ph.D. students will use in their research. It starts with an intensive week-long course followed by weekly Methods Brown Bag sessions (more details will be provided on the last day of the camp). This course will reintroduce students to basic topics in probability, linear algebra, and calculus through lectures, problem sets, and some homework assignments. Students will also learn to employ the statistical software packages and gain exposure to data and workflow management through a series of labs.

Week Schedule

Time	Mon 8/17	Tue 8/18	Wed 8/19	Thu 8/20	Fri 8/21
9:00 – 11:00	Talking in Math	Probability	Calculus	Linear Algebra	Advanced Topics
11:00 – 11:10	Break	Break	Break	Break	Break
11:10 – 1:30	Talking in Math	Probability	Calculus	Linear Algebra	Advanced Topics
1:30 – 2:30	Lunch	Lunch	Lunch	Lunch	Lunch
2:30 – 4:00	Setup	Basics	Exploration	Mission	AI
4:00 – 4:30					Open Discussion

Daily Topics

Day 1 — Talking in Mathematics / Setup

Math

- Theory, concept, measure, variable, and hypothesis
- Sets and set notation: unions, intersections, subsets, sample spaces
- Functions: domain, codomain, and range
- Summation and product notation
- Exponents, logarithms, and roots
- Solving equations and finding roots

Programming — Setup

- Setup: R, RStudio, Zotero, GitHub
- Basic introduction to R and RStudio: interface, scripts
- Basic practices: commenting, code chunks

Day 2 — Probability / Basics

Math

- Basic probability
- Compound probability
- Random variables and expectation algebra
- Conditional expectation and the type-shift ($E[Y|X = x]$ vs. $E[Y|X]$)

Programming — Basics

- Why programming or coding
- R basics
- Objects, data types, and data structures (vectors, matrices, lists)
- Exercise: understand a real replication dataset

Day 3 — Calculus / Exploration

Math

- Limits
- Differentiation and optimization (first-order conditions)
- Integration basics
- Area under the curve

Programming — Exploration

- Data structures continued: factors and data frames
- Conditionals in base R
- Descriptive statistics and aggregation
- Exercise: explore the summary/descriptive statistics of the sample dataset

Day 4 — Linear Algebra / Mission

Math

- Matrix readings
- Transformations
- Sampling distributions
- Basic derivations

Programming — Mission

- Housekeeping: working directories, importing and exporting datasets, packages, YAML, code chunks
- Data wrangling with `dplyr` and `tidyr`
- Grammar of graphics with `ggplot`
- Why functions

Day 5 — Intro to Advanced Topics / AI

Math

- Sampling distributions; the law of large numbers and the central limit theorem, by demonstration
- Frequentist vs. Bayesian inference
- Causality and causal inference
- The road ahead: where this week's tools reappear in first-year coursework

Programming — AI

- Coding with AI

References

- Moore, W. H., & Siegel, D. A. (2013). *A Mathematics Course for Political and Social Research*. Princeton University Press.
- **Seeing Theory — Basic Probability** (Brown University) (seeing-theory.brown.edu/doc/basic-probability.pdf).
- **Harvard IQSS Prefresher — Probability** (iqss.github.io/prefresher/06_probability.html).